

Jiangtao Xie

18650639138 | xiejiangtao@stu.ouc.edu.cn

Educational background

➤ Ocean University of China (A-Class of "Double First-Class" Institution, 985 Institution)

Department of Information Science and Engineering

Electronic Information Engineering Major

➤ Overall academic performance and ranking: 84.68/100 (rank:24/111)

IELTS : 6.5

➤ Course grade:

High-resolution spectral analysis (97) 、 Deep learning (95.6) 、 Advanced language programme design (93) 、

Electronic circuit design (95) 、 Fundamentals of Circuit Analysis (91) 、 Artificial intelligence (91.5) 、

Electronic Skills Training (95) 、 Quantum mechanics and quantum information theory (90) 、

Method of mathematical physics (91.6) 、 Computer aided drafting (90)

High School Academic Competitions : CHSBO 2022 Provincial First Prize | CPho 2022 Provincial Second Prize

Scientific research experience

Design of a ground-to-air collaborative inspection robot for chemical plants with active suspension system
Provincial-level Innovation Project

Mechanical Structure Design: Designed an electric-cylinder-driven quadruped robot chassis and completed 3D modeling and motion simulation in SolidWorks.

Active Suspension Control: Developed a cascade PID control system based on MPU6050 attitude feedback, enabling the robot to maintain stability on 15° slopes and overcome 20 cm obstacles, with body inclination kept below 3°. **Electronic Control System:** Designed the main control board based on the STM32F103VET6, incorporating power management and CAN, SPI, and I²C communication interfaces.

Multi-Sensor Integration: Integrated the MQ-2 gas sensor, SHT30 temperature and humidity sensor, and RPLIDAR A1 LiDAR for environmental monitoring and perception.

High-Speed Data Acquisition: Designed a custom PCB and signal-conditioning circuits to achieve multi-source sensor data acquisition at a sampling rate of at least 100 Hz.

Autonomous Navigation and Obstacle Avoidance: Implemented SLAM-based localization, A* path planning, and Dynamic Window Approach navigation on the ROS platform, achieving a positioning error within ±10 cm.

Ground – Air Collaborative Positioning: Optimized the fusion of visual sensing and UWB positioning to enable precise docking between the UAV and the ground robot, with a docking error of less than 3 cm.

Numerical simulation of sedimentary sequence evolution and hydrocarbon migration and accumulation model under the control of strike-slip pull-apart basin formation and inversion structure

University-level Innovation and Entrepreneurship Program

Multiphysics Algorithm Development: Independently developed a three-dimensional two-way sedimentary – tectonic coupling framework by integrating ASPECT and Badlands, demonstrating strong capabilities in numerical algorithm design, multiphysics coupling, and scientific software development.

Finite-Element Modeling: Built a visco-elastic-plastic finite-element model incorporating dynamic boundary conditions, realistic crustal thermal structures, fault geometry, and material heterogeneity.

High-Performance Numerical Simulation: Implemented full-lifecycle simulations of the dynamic feedback between deep tectonic deformation and shallow sedimentary loading, with experience in large-scale computational modeling, parameter configuration, convergence analysis, and simulation stability control. **Geodynamic**

Data Analysis: Analyzed stress migration, deformation patterns, sedimentary evolution, and fault-controlled basin development, demonstrating strong skills in extracting geological mechanisms from complex simulation outputs.

Scientific Visualization: Processed and visualized three-dimensional simulation results, including stress fields, velocity fields, deformation patterns, sediment thickness, and basin morphology.

Machine Learning Integration: Applied machine learning together with coupled numerical simulation to develop a paleotopographic inversion method for marginal-sea hydrocarbon basins. One patent application has been approved.

Geoscientific AI and Inversion: Contributed to two granted invention patents on multi-source cross-modal geostress prediction and machine-learning-based paleotopographic inversion, with experience in geological data fusion, intelligent modeling, inverse-problem solving, and computational geoscience.

Underwater cleaning robot for ship propellers

University-level Innovation and Entrepreneurship Program

Electronic System Integration: Led the design of a Jetson Nano-based distributed control system and independently developed a multi-channel power management PCB, ensuring stable power supply, low-noise operation, and reliable integration of complex underwater robotic hardware.

Deep Learning-Based Vision Development: Developed a real-time underwater image enhancement model to address turbidity, color distortion, uneven illumination, and cavitation-jet interference.

Edge AI Deployment: Optimized and lightweighted the model for real-time high-definition video processing on Jetson Nano, demonstrating strong capabilities in embedded AI, model optimization, and hardware – software co-design.

System Performance Improvement: Significantly enhanced underwater visual perception and provided reliable technical support for remote monitoring and precise robot control.

Fabrication and Application of Visible-Transparent Infrared Organic Photodetectors (OPDs)

Research Internship at the University of Science and Technology of China

Device Research and Optimization: Participated in Prof. Jun Liu's resonant boron-nitrogen bond-based SWIR detector research; studied 1000-3000 nm photoelectric response and optimized fabrication to retain transparency and improve infrared sensitivity.

Image Processing and Device Integration: Processed infrared imaging data, evaluated image quality, and contributed to integrating flexible, visible-transparent organic photodetectors with image acquisition and processing.

Metagenomic analysis reveals Northwest Pacific Ocean as a reservoir and evolutionary hub of antibiotic resistance genes

Environmental Pollution | SCI JCR Q1 | Third Author

Bioinformatics Pipeline: Integrated Trimmomatic, MEGAHIT, Prodigal and DIAMOND/CARD for quality control, assembly, gene prediction and antibiotic resistance gene annotation.

Statistical Modeling: Built Spearman co-occurrence networks with bootstrap validation; applied zero replacement and centered log-ratio transformation to address compositional bias.

Functional Analysis: Used KEGG/eggNOG annotation to characterize efflux-pump-mediated resistance and interpret marine resistance gene patterns.

Tensor-Product Quantum Convolutional Neural Networks for Image Classification

Architecture and Modeling: Contributed to a two-layer, multichannel QCNN using amplitude encoding and tensor-product parameter sharing, with complementary Z/X readout and classical pooling/re-encoding.

Simulation and Evaluation: The project achieved 98.36%, 87.53% and 85.84% accuracy on MNIST, FashionMNIST and PneumoniaMNIST, respectively, in state-vector simulations across three random seeds.

Compression and Robustness: Reduced the MNIST model from 3,640 to 520 parameters (85.71%) while retaining 96.06% accuracy; evaluated trainability, readout ablations and gate-level noise robustness.

Subject competition experience

National Undergraduate Competition on Embedded Chips and Systems Design (National Third Prize)

AI Algorithm Design: Developed a hybrid recognition architecture combining a lightweight LeNet neural network with font-matching algorithms to accurately recognize Chinese characters, letters, and numbers.

Hardware Implementation: Used Verilog HDL to complete the full hardware design process, including image preprocessing, license plate localization, character segmentation, and recognition.

Hardware – Software Co-Design: Efficiently mapped the recognition algorithms onto FPGA through modular architecture and pipelining, demonstrating strong capabilities in AI algorithm optimization, digital circuit design, and embedded system integration.

Performance Optimization: Improved real-time processing capability and energy efficiency, enabling reliable deployment of the complete license plate recognition pipeline on resource-constrained FPGA hardware.

International Genetically Engineered Machine Competition (Global Silver Award, Hardware Compliance Gold Award)

Robotic System Development: Built a ROS- and Gazebo-based digital twin for wound contour recognition, manipulator motion, and 3D path planning, and developed a custom inverse-kinematics solver for closed-loop simulation and physical-system deployment.

Two-Stage Visual Perception: Developed a coarse-to-fine wound recognition pipeline in which YOLOv8-seg first rapidly localizes the wound region, followed by FCN-based fine-grained segmentation within the detected area to improve boundary accuracy, detail preservation, and robustness across different skin tones.

AI Agent and Computational Modeling: Integrated AI agents into the dry-lab workflow for literature retrieval, antimicrobial peptide generation, property prediction, explainable ranking, result comparison, and experimental data interpretation, connecting computational screening with wet-lab validation. Conducted **molecular dynamics simulations** to quantitatively analyze hydrogel gelation mechanisms and peptide– carrier interactions

Multimodal System Integration: Combined polar-coordinate path planning, motion-control optimization, low-cost depth estimation, temperature and pH sensing, and IoT transmission, demonstrating strong capabilities in robotics, computer vision, AI modeling, automation, and hardware – software integration.

International Directed Evolution Contest (Best Molecular Evolution Results | Best Software Design | Best Machine Learning Driven Directed Program Nomination)

Analysis: Analyzed enzyme structures and key residues to guide mutation-site selection and candidate design.

Modeling: Built machine learning models to predict and rank promising enzyme variants using multidimensional amino-acid features.

Validation: Integrated computational screening with protein expression, purification, and activity evaluation to validate optimized mutants.

CCF Quantum Computing Programming Challenge - Original Source Cup (National Second Prize)

Quantum Theoretical Modeling: Derived Hamiltonian matrices, parameterized quantum states, energy expectation functions, and optimal variational parameters, transforming quantum-mechanical formulations into executable optimization models.

Quantum Circuit Implementation: Implemented quantum algorithms in PyQPanda3, including entanglement preparation, quantum teleportation, measurement-based control, conditional correction, probability analysis, and fidelity verification.

Variational Quantum Architecture: Designed an 8-qubit baseline VQC and a lightweight 2-qubit Data Re-uploading classifier, enabling complete encoding of 16-dimensional medical features with only 6 trainable quantum parameters.

Distillation and Compression: Integrated knowledge distillation, loss regularization, circuit compression, and hyperparameter optimization, reducing qubit usage by 75% and trainable quantum parameters by 92.5% while maintaining effective classification performance.

National College Students' Mathematical Modeling Contest (Provincial Second Prize)

National College Students' Intelligent Vehicle Competition (Provincial Second Prize)

China College Students ' Engineering Practice and Innovation Ability Competition (Provincial Third Prize)

Mathematical Contest In Modeling 2026 Honorable Mention

Other information

➤ Engineering ability : skilled use of Solidwork, Jialichuang EDA, MATLAB, Fushion360, Dify, Coze, Linux system, supercomputing, GPU cluster use, training experience, large model deployment fine-tuning experience, use ROS system to control the mainstream manipulator, mainstream single-chip microcomputer use experience.